Personalized Oral Hygiene Framework for Longitudinal Toothbrushing Monitoring in-the-wild

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Abstract

Poor oral hygiene is a leading cause of dental caries, gingivitis, and periodontitis, yet toothbrushing quality remains largely unmonitored outside clinical settings. Wearable sensing offers an opportunity to continuously assess brushing behaviors and support both individuals and clinicians through longitudinal monitoring. This paper presents *O-Phar*, an AI-driven framework for personalized oral hygiene monitoring using smartwatch-based inertial measurement unit (IMU) sensing. The framework recognizes brushing motions (circular, horizontal, and vertical) and sextant-level oral coverage from wrist kinematics using lightweight deep learning models deployed on smartphones. Beyond presenting the framework, we discuss the key challenges encountered while developing a longitudinal toothbrushing monitoring system for real-world deployment. Specifically, we identify four major challenges: (1) substantial inter-user variability in brushing biomechanics and IMU signatures that limits the performance of generalized models; (2) longitudinal feature drift, where IMU feature distributions for the same user evolve across brushing sessions, reducing the effectiveness of one-time personalization; (3) the scarcity of accurately labeled IMU data, which hinders continual model adaptation and motivates cross-modal video-to-IMU annotation; and (4) the difficulty of developing reliable longitudinal models that correlate brushing behaviors with clinical oral health measures such as plaque score. We describe our proposed solutions, summarize the insights gained from real-world data collection, and discuss the personalization of longitudinal oral health monitoring.

1 Introduction

Maintaining good oral hygiene is essential for both oral and systemic health, as emphasized by the American Dental Association (ADA). However, many individuals lack self-awareness regarding their toothbrushing habits, including the brushing techniques they employ and the amount of time spent cleaning different oral regions [5]. As a result, individuals often overbrush certain areas while neglecting others or repeatedly perform ineffective brushing motions, such as horizontal scrubbing, circular brushing, or vertical strokes, without realizing whether these motions are clinically appropriate. In addition, proper brushing technique is critical for removing plaque along the gingival margin and preventing oral diseases such as gingivitis and periodontitis, which have also been linked to systemic conditions, including cardiovascular disease, stroke, and diabetes [10].

Commercial smart toothbrushes developed by companies such as Oral-B and Philips use proprietary hardware [9], making them expensive and limiting their accessibility for routine home monitoring. More recent research systems, including ToMoBrush, SmarTeeth, and AI-assisted dental imaging systems [11, 13, 14], focus on detecting tartar, gingival inflammation, or dental caries after these conditions have already developed. Consequently, they provide diagnostic assistance rather than monitoring daily brushing behaviors, which ultimately contribute to these oral diseases.

Advances in wearable sensing and human activity recognition have shown the potential of smartwatches for monitoring toothbrushing behavior. Prior work [8, 11] has shown that wrist-worn IMU sensors can capture wrist kinematics to infer the oral region being brushed.



Figure 1: *O-Phar's* application framework

By utilizing accelerometer, gyroscope, and magnetometer measurements, these systems estimate wrist orientation through roll, pitch, and yaw angles to determine whether users follow the ADA-recommended Modified Bass brushing technique [9]. More recent systems, including mORAL [1] and Hygiea+ [12], further perform fine-grained recognition of brushing activities across multiple oral surfaces, including the roof of the mouth and tongue, demonstrating the feasibility of wearable sensing for detailed oral hygiene monitoring.

Despite these advances, existing systems have several important limitations. First, they are primarily evaluated under controlled laboratory settings and rely heavily on manually annotated IMU data for training and evaluation. Second, most prior work focuses on determining whether users follow the Modified Bass technique or identifying the oral region being brushed, without explicitly recognizing the brushing motion

itself (e.g., circular, horizontal, or vertical). Finally, these systems are designed for short-term activity recognition and do not investigate how brushing behaviors evolve over time, making them unsuitable for longitudinal monitoring. Longitudinal toothbrushing monitoring enables dentists to objectively assess patient adherence to prescribed brushing techniques, particularly for individuals with gingivitis or periodontitis, and supports caregivers in monitoring oral hygiene behaviors of older adults and patients with dementia. Furthermore, continuous monitoring establishes the foundation for predicting clinically relevant oral health indicators, such as plaque score and gingival index, enabling earlier intervention before disease progression.

Motivated by these clinical needs and in collaboration with periodontics experts at the University of Maryland School of Dentistry, we developed *O-Phar*, a personalized smartwatch-based framework that recognizes toothbrushing motions (circular, horizontal, and vertical) and oral regions (six oral sextants) using wrist-worn IMU sensing (Fig. 1). During its development, we encountered several challenges that affect longitudinal toothbrushing monitoring.

① *Inter-user feature variability:* IMU feature distributions vary considerably across individuals because of differences in brushing style, wrist biomechanics, smartwatch placement, and natural brushing habits. Consequently, a generalized model trained across multiple users often fails to accurately capture user-specific brushing behaviors.

② *IMU ground-truth annotation:* Raw IMU signals contain little semantic information, making it extremely difficult to manually annotate brushing motions and oral regions directly from sensor data. Accurate annotation therefore requires an additional sensing modality, such as synchronized video, to provide reliable ground truth.

③ *Longitudinal oral health monitoring:* Our long-term objective is to correlate longitudinal brushing behaviors with clinically obtained plaque and gingival scores [4]. Achieving this objective requires models that remain robust despite temporal changes in brushing style, wrist orientation, and sensor placement. Developing such longitudinally invariant models remains a significant open challenge.

To address these challenges, we explored several complementary techniques.

① **Personalized models:** Rather than relying solely on a generalized model, we investigate lightweight personalization strategies that adapt the learned feature representation to individual users using a small amount of calibration data.

② **Cross-modal video-to-IMU annotation:** Since brushing actions are easier to annotate in videos than raw IMU signals, we collect synchronized videos during training, manually annotate them, and transfer the labels to IMU data via cross-modal distillation. We will open-source the annotation tool upon acceptance to support reproducible research.

③ **Longitudinal data collection pipeline:** We develop an integrated watchOS/iOS application that synchronously records IMU data and videos during training. Data are collected over multiple weeks in both controlled and uncontrolled settings. In controlled sessions, participants perform predefined brushing motions or brush specific oral regions for fixed durations to obtain reliable ground truth. In uncontrolled sessions, participants brush naturally during their daily routines without receiving any instructions regarding brushing motions or oral regions. We will open-source the watchOS/iOS application upon acceptance.

We evaluate *O-Phar* on 16 participants across multiple weeks of data collection. Depending on user-specific variability and adaptability, the personalized model improves oral region classification accuracy by **17.1 to 34.5 percentage points** and brushing motion recognition accuracy by **8.1 to 28.4 percentage points** compared to the generalized LSTM model. Despite these improvements, we observe that the accuracy of the personalized model gradually degrades over time, even under controlled brushing conditions, indicating that one-time personalization is insufficient for longitudinal deployment. Moreover, models trained using controlled brushing data generalize poorly to natural daily brushing sessions, achieving an accuracy of only approximately **23.4%** on region classification. These observations show that although techniques such as LSTMs, personalization, and cross-modal distillation improve short-term performance, they remain inadequate for robust longitudinal toothbrushing monitoring.

2 Data Collection and Annotation Tool

2.1 Hardware, Software, and Data Collection

We developed a custom watchOS/iOS data-collection pipeline for *O-Phar*, in which participants wore a commodity Apple Watch on their dominant wrist during brushing. The watch recorded 9-axis IMU data at 100 Hz, including accelerometer, gyroscope, magnetometer, and fused attitude signals, while the paired iPhone recorded the video data and synchronized video and IMU recordings via a unified Start/Stop protocol, ensuring modality alignment to within 200 ms. Video was collected during training to provide a visual reference for annotation; however, the video modality is not required during inference, where the system operates solely on IMU data. Data collection was conducted in two phases. In Phase 1, we recruited 10 participants and collected two controlled sessions per participant. In each controlled session, participants were instructed to perform each brushing motion and each oral region for 10 seconds in a predefined order, making the transition boundaries relatively clear. In Phase 2, we recruited 6 participants for longitudinal data collection over five consecutive weeks. Participants provided controlled data once per week using the same instructed protocol and additionally recorded natural, uncontrolled brushing sessions across

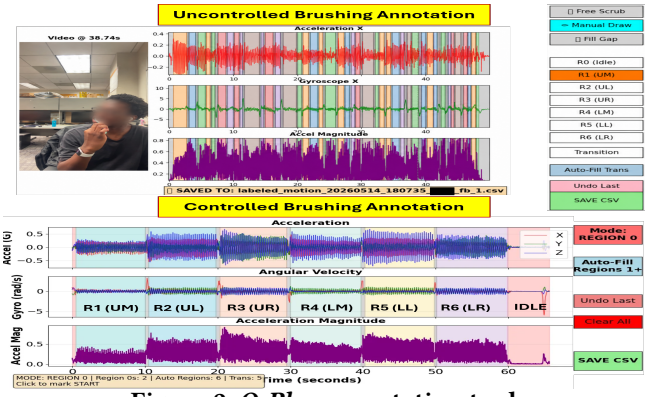


Figure 2: *O-Phar* annotation tool

the remaining days of each week. The Phase 2 longitudinal protocol was designed to capture both structured and naturalistic toothbrushing behavior. Participants recorded either morning or evening brushing sessions based on convenience, without enforcing a fixed time of day.

2.2 Annotation Pipeline

To annotate the collected data, we developed an interactive visual-kinematic annotation tool that synchronizes the video timeline with smartwatch IMU plots. The tool enables annotators to inspect the video alongside accelerometer and gyroscope streams and assign labels for oral region, brushing motion, transition, and idle segments. Controlled sessions were easier to annotate because users followed a predefined order and fixed-duration instructions, whereas uncontrolled sessions required careful visual inspection due to arbitrary brushing order, variable segment durations, and natural movement patterns. This annotation pipeline provides labeled IMU windows for training and analyzing both controlled and in-the-wild brushing behaviors. Fig. 2 illustrates the custom annotation tool developed to label synchronized video and raw IMU data for both controlled and naturalistic brushing sessions across all six oral regions. In the controlled setting, transitions between instructed regions and motions are relatively clear because participants follow a predefined brushing order and duration. In contrast, uncontrolled brushing exhibits irregular motion patterns, variable brushing durations, and ambiguous transitions between regions, making annotation substantially more challenging. These differences highlight the practical difficulties of deploying toothbrushing-monitoring frameworks in the wild.

3 Preliminary Observations

To better understand the structure of the collected IMU data, we extracted the features described in Section 4 and analyzed their distributions across brushing motions, oral regions, and users. This preliminary analysis revealed noticeable overlap between neighboring oral regions and between certain motion classes, particularly horizontal and vertical brushing. A

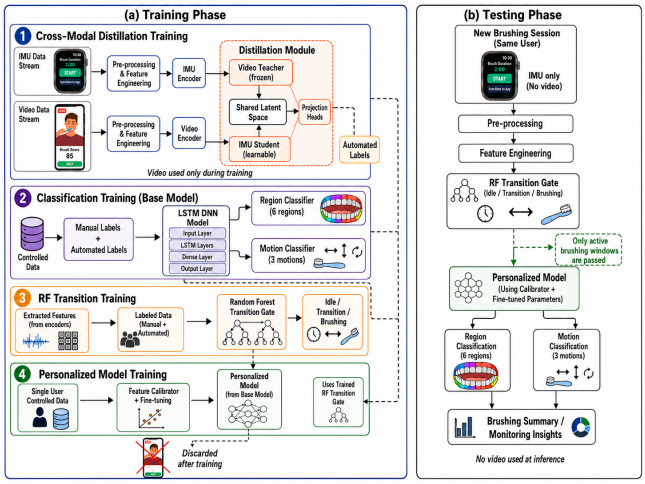


Figure 4: *O-Phar* system overview

key observation was that several features exhibited stronger separation across users than across brushing classes, indicating that person-specific signal characteristics dominate the learned feature space. Fig. 3 demonstrates this effect using a representative IMU feature. The minimum Delta-band (0.5–4 Hz) power of the attitude roll angle. The density distributions show that samples cluster more distinctly by user identity than by brushing region, revealing significant inter-user variability caused by differences in brushing style, wrist biomechanics, and smartwatch placement. This suggests that a generalized model must first compensate for user-specific feature shifts before it can reliably learn brushing semantics. These observations motivate the need for personalization, where a generalized model is fine-tuned or calibrated using user-specific data to better capture an individual's brushing signature.

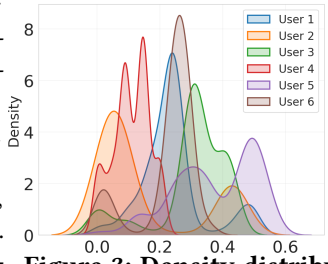


Figure 3: Density distribution of a feature across users

4 *O-Phar's* Framework Overview

In this section, we present *O-Phar*, a personalized toothbrushing activity recognition framework for identifying brushing motions and oral regions during regular brushing sessions, as shown in Fig. 4. The framework includes four modules: cross-modal video-to-IMU label annotation, an LSTM-based generalized classifier for three motions and six regions, a Random Forest transition detector, and a personalization module.

4.1 Cross-Modal Distillation and Automatic Label Generation

Manual annotation of brushing regions and motions does not scale to longitudinal uncontrolled brushing. We therefore use an offline cross-modal distillation pipeline, inspired by

COMODO [2], where synchronized smartphone video temporarily supervises smartwatch IMU during training. Video is used only for label generation and representation transfer; at deployment, inference relies solely on IMU data.

During training, the video and IMU streams are temporally aligned by correcting iPhone–Apple Watch clock drift to within 200 ms using synchronized filename timestamps. Each 300 ms IMU window is then matched to the nearest corresponding video frame range, and unmatched windows are discarded. We train a video teacher, inspired by TimeSformer [3], to capture visual brushing cues by localizing the brush and mouth region and pooling ROI features. The IMU student, implemented using our LSTM model, is trained to match the teacher's latent relational structure rather than directly copying class probabilities.

To improve recognition accuracy, the student is optimized using four complementary losses: (i) a relational KL alignment loss that matches temperature-scaled similarity distributions over Q, enabling the IMU model to preserve the teacher's latent similarity geometry; (ii) a pairwise projection alignment loss that maximizes cosine similarity between matched video–IMU embeddings; (iii) a center loss that pulls steady-state region windows toward learned class centroids to improve intra-class compactness; and (iv) a transition-boundary loss that explicitly models region hand-offs, preserving the transitional patterns between oral regions. We further use a specialist ensemble with temperature $T = 20$ to improve discrimination among commonly confused neighboring regions, such as the upper-left and lower-left surfaces. At inference, the video teacher, projection heads, and queue are discarded, and only the lightweight IMU student, transition gate, and personalized region/motion LSTMs are used on-device.

4.2 Toothbrushing Classification Pipeline

The *O-Phar* classification pipeline consists of four main stages: (i) IMU preprocessing and feature extraction, (ii) training generalized classifiers for six oral regions and three brushing motions, (iii) detecting idle and transition intervals using a Random Forest gate, and (iv) adapting the generalized model to each user through lightweight personalization.

Preprocessing and Feature Extraction: The smartwatch records 9-axis IMU streams, including acceleration, angular velocity, magnetic field, and fused attitude angles (roll, pitch, and yaw). We first correct clock drift to within 200 ms and align IMU windows using shared filename timestamps (`YYYYMMDD_HHMMSS`). The signals are segmented into 300 ms windows with 50% overlap, which is sufficient to capture stroke-scale brushing dynamics around 3–4 Hz. We apply a 0.5–10 Hz bandpass filter, followed by vertical alignment and user-coordinate-system mapping using the magnetometer-derived facing angle [12]. For each window, we extract time-domain statistics, including mean, standard deviation, skewness, kurtosis, and zero-crossing rate; spectral features, including STFT centroid, entropy, flux, and band power near 2 Hz; and jerk-based features that capture rapid brushing motion. Random Forest feature importance with cross-validation is then used to reduce the 4,900+ candidate features into task-specific subsets of approximately 170 features for region, motion, and transition classification.

Generalized Classification Model: *O-Phar* uses a two-stage classification pipeline. First, a lightweight Level-1 Random Forest gate classifies each window as idle, transition, or active using jerk and spectral features, filtering out non-brushing intervals before region and motion recognition [7]. Active windows are then passed to two independent LSTM classifiers: one for six oral regions (upper left, upper middle, upper right, lower left, lower middle, and lower right) and one for three brushing motions (circular, horizontal, and vertical). We train separate models for the two tasks because region and motion recognition rely on different signal characteristics, and separating them avoids negative transfer. Each LSTM consists of two layers and processes sequences of $T = 40$ windows, corresponding to approximately 4 s of temporal context. The hidden dimension is set to $d = 128$, with dropout $p = 0.3$ applied to reduce overfitting on the limited training data.

Personalized Fine-tuning: To address the inter-user feature variability, we use subject-adaptive transfer learning [6]. The generalized model consists of a shared LSTM encoder, a lightweight per-user **Feature Calibrator** with affine scale and bias parameters, and task-specific classification heads for region and motion recognition. During base training, the generalized LSTM model is trained using labeled data from Phase 1 participants. During personalization, the LSTM encoder is frozen, and only the Feature Calibrator and classification head are fine-tuned using labeled controlled sessions from the target user. This strategy requires substantially less labeled data than training a user-specific model from scratch and remains lightweight.

Personalization improves region and motion recognition by aligning user-specific IMU statistics with the feature manifold learned by the generalized model, a strategy commonly used in wearable activity recognition and toothbrushing monitoring systems [1, 9, 12]. The Level-1 transition gate remains a shared Random Forest classifier trained on controlled data and is not personalized in the current system.

5 Results and Insights

5.1 Controlled Sessions

As described in Section 2.1, Phase 1 data trains the cross-modal distillation module, generalized LSTM, and shared Random

Forest transition gate, while Phase 2 data personalizes the LSTM for each longitudinal user. We first evaluate the framework under controlled brushing conditions.

Generalized vs. Personalized Model: The generalized base model achieves 51.9%±4.1% region accuracy and 63.6%± 14.9% motion accuracy across the six Phase 2 users. Per-user base accuracy ranges from 48.9% to 54.8% for region classification and from 53.1% to 74.1% for motion classification. These results reflect the substantial inter-user feature variability, confirming that a global model alone is insufficient for reliable toothbrushing recognition. Fig. 5 show the effect of personalization for region classification. After fine-tuning with approximately one day of calibration data, mean region accuracy improves from 51.9% to 79.3% ± 9.1%, while mean motion accuracy increases from 63.6% to 82.1%. Across users, personalization provides gains of +18.1 to +36.9 percentage points for region classification and +8.0 to +29.0 percentage points for motion classification. The post-personalization confusion matrices further show that region classification accuracy varies from 68% to 96% across the six oral regions, while motion recognition accuracy ranges from 68% to 90% across the three brushing motions. Interestingly, the confusion matrices are asymmetric, indicating that misclassification patterns are directional rather than reciprocal.

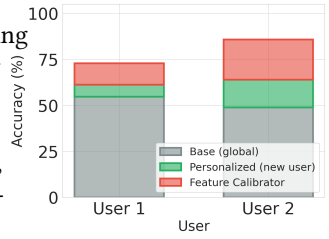


Figure 5: Region classification accuracy

We also observe that additional calibration data improves personalization. Region and motion accuracy increase steadily as more user-specific samples are used for fine-tuning, eventually converging near 79% for region classification and 82% for motion classification.

The **Random Forest gate classifier** is evaluated on six users. It achieves a region-switch detection rate of 95%, and a false-alarm rate of 4% across idle, transition, and active classes, showing reliable detection of non-brushing intervals before region and motion classification.

Cross-Modal Distillation Framework: We evaluate the cross-modal distillation module using N=2509 synchronized video–IMU windows from controlled sessions, where each IMU segment is 300 ms with 50% overlap. Distillation quality is assessed using video–IMU pairing correctness. The matched video–IMU embeddings achieve a cosine similarity of 0.995 ($\sigma = 0.019$), compared to 0.198 ($\sigma = 0.011$) for randomly permuted pairs, yielding a similarity gap of 0.797. Per-region matched similarity ranges from 0.993 for IDLE to 0.997 for brushing regions, indicating that the IMU student consistently captures the video teacher's latent structure across classes. The student is trained using a combination of relational KL alignment over a warmed FIFO queue Q, pairwise projection alignment, center loss on steady-state windows, and boundary alignment at region transitions. At convergence, the alignment loss reaches 0.014, with transition samples comprising 15.1% of Q, suggesting that the model captures soft boundaries between neighboring oral regions. The model also shows strong cross-session generalization for selected users, achieving accuracies of 92.4% and 88.4% on unseen sessions. On the test set, CKA between video and IMU embeddings is 0.42, while mean CCA correlation reaches 0.97, confirming strong correspondence between the two modalities. Correlation-guided rules further place all seven region classes in the auto-export eligibility set using a ≥ 0.75 matched-pair correlation threshold. Qualitative results in Fig. 7 show that the student's predictions closely follow the ground-truth labels for the controlled setting.

5.2 Deployment In-the-Wild

Although the controlled experiments show promising results, deploying *O-Phar* in naturalistic settings remains challenging. Our goal is to support in-the-wild toothbrushing monitoring with minimal supervision, where video is not collected during deployment for privacy reasons and IMU data must be labeled or interpreted remotely through the cloud. However, our Phase 2 longitudinal study reveals three major challenges: (i) intra-user temporal feature variability, (ii) limited transfer of controlled cross-modal distillation models to natural brushing sessions, and (iii) user-dependent personalization behavior, raising the open question of how much calibration data is needed before a personalized model becomes stable.

Intra-user Temporal Variability: In addition to the inter-user variability discussed earlier, we observe substantial temporal variability within the same user. Although the brushing action remains the same, the feature distribution gradually shifts across sessions, reflecting changes in wrist orientation, brushing style, smartwatch placement, and natural behavioral variation. Overall, uncontrolled brushing exhibits stronger temporal feature drift than controlled brushing, with a Week 1–Week 4 increase of 95.6% compared to 51.2% in the controlled setting. The largest divergence occurs where uncontrolled variability increases by 43.3% compared to only 8.2% under controlled brushing, suggesting that natural brushing behavior accelerates feature drift beyond what is observed in structured calibration sessions.

User-dependent Personalization: Personalized fine-tuning improves performance in controlled settings, but the amount of data required for adaptation varies across users. As shown in Fig. 6, some users reach peak accuracy with relatively few samples, whereas others show limited improvement even with additional calibration data. For example, User 3 reaches peak performance with approximately 500 samples, while User 4

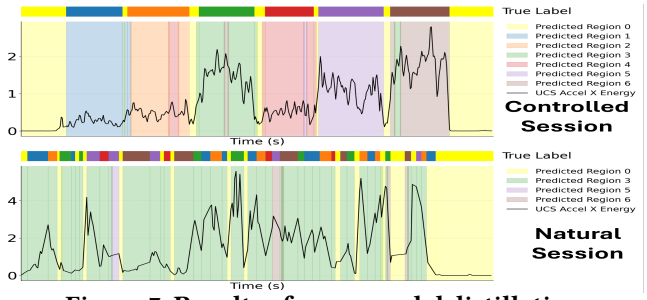


Figure 7: Results of cross-modal distillation

remains around 86% accuracy with limited additional gains. In controlled brushing, personalization improves User 1's motion accuracy from 0.530 to 0.814 and region accuracy from 0.548 to 0.719. Similarly, User 2 improves from 0.741 to 0.822 for motion and from 0.489 to 0.834 for region classification. However, this improvement does not fully transfer to uncontrolled brushing, where personalized region classification accuracy drops to 0.234. This indicates that controlled calibration data does not capture the full variability of natural brushing, including arbitrary region transitions, unconstrained wrist motion, and inconsistent brushing order.

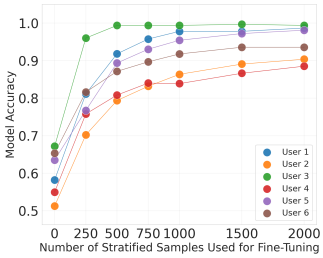


Figure 6: Personalized fine-tuning across days

Cross-modal Distillation under Natural Behavior: As shown in Fig. 7, the cross-modal distillation model struggles to annotate IMU data accurately in uncontrolled sessions. This degradation is caused by temporal drift and changes in brushing behavior that are not sufficiently represented in controlled training data. The standard deviation of a representative feature increases over time in both settings, but the increase is much larger during uncontrolled brushing: from 0.090 in Week 1 to 0.176 in Week 4. In contrast, the controlled setting remains stable during the first two weeks (0.086–0.085) and increases moderately after the gap week, reaching 0.130 by Week 4. These results show that controlled protocols reduce variability but do not eliminate longitudinal drift.

Future Work: Collectively, these observations explain why models trained under structured laboratory conditions often fail to generalize to daily brushing. Future deployments will require continual adaptation, stronger domain generalization, and potentially additional sensing modalities, such as audio, to improve robustness in naturalistic settings. Our experience suggests that future systems will require continual adaptation, self-supervised representation learning, automated video annotation, and more fine-grained oral region recognition, including lingual and palatal tooth surfaces, to enable reliable deployment in real-world settings.

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